

Efficient Semantic Segmentation: Leveraging Knowledge Distillation for Modality Reduction

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Abstract—This study aims to improve the efficiency of urban vegetation segmentation by transferring knowledge from a multi-modal model into a simplified single-modal model. Accurate vegetation segmentation is critical for urban planning and environmental management but often relies on complex data sources such as RGB, near-infrared (NIR), and LiDAR. To address this, we first trained a multi-modal U-Net semantic segmentation model using RGB, NIR, and LiDAR data. Subsequently, we utilized knowledge distillation (KD) to transfer the multi-modal insights of this teacher model into a more compact RGB-only student model. We evaluated the models on an urban vegetation dataset, achieving a mean Intersection over Union (mIoU) of 0.7132 with the multi-modal teacher and 0.6002 with the distilled RGB-only student, significantly surpassing RGB-only (0.4091 mIoU) and NIR-only (0.5044 mIoU) baselines. The student notably retained high performance in challenging categories such as Street Trees and Shrubs. Our findings demonstrate that KD effectively reduces the modality requirement for segmentation tasks, simplifying deployment with only a modest compromise in accuracy.

Keywords—knowledge distillation, semantic segmentation, multi-modal fusion, remote sensing, U-Net, modality reduction

I. INTRODUCTION

Remote sensing-based semantic segmentation of vegetation matters in environmental monitoring, forestry, and urban planning. It allows researchers to map deforestation and ecosystem health, track land-cover changes, and measure urban green spaces. In precision agriculture, semantic segmentation differentiates crops from weeds for automated control systems [1]. Recent work demonstrates the ability to generate precise, pixel-level vegetation maps from aerial and satellite imagery, providing critical data for sustainable development and resource management [2].

State-of-the-art segmentation methods combine multiple data sources to exploit complementary spectral and structural information [3]. Although this fusion boosts accuracy, collecting and processing all modalities is expensive and complex. To bridge the gap between high accuracy and efficient deployment, we train a multi-modal U-Net (teacher) on RGB, NIR, and LiDAR and then distill its knowledge into a smaller, RGB-only U-Net (student). The distilled student surpasses a single-modal baseline while requiring only RGB inputs.

In this work, our contributions are threefold: (1) we propose a cross-modal distillation framework that eliminates extra inputs for urban vegetation segmentation; (2) we provide a quantitative comparison between the multi-modal teacher, the distilled RGB-only student, and two single-modal baselines; and (3) we show that the distilled student recovers most of the teacher's performance with minimal accuracy loss.

II. BACKGROUND AND RELATED WORK

A. Deep Learning and Multi-Modal Fusion for Vegetation Segmentation

Convolutional neural networks (CNNs) have transformed remote sensing image analysis, enabling precise pixel-level vegetation mapping from aerial and satellite imagery. The U-Net architecture [4], with its encoder-decoder design and skip connections, has become a dominant choice for vegetation segmentation tasks [5]. The encoder compresses contextual information while the decoder restores spatial resolution, and skip connections bridge the two to recover fine details such as individual trees or crop rows. This structure allows accurate delineation of vegetation classes. Fig. 1 shows the standard U-Net architecture, highlighting the encoder-decoder paths and skip connections.

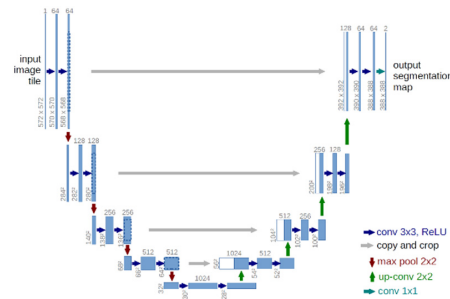


Fig. 1. Schematic diagram of U-Net architecture showing encoder-decoder path with skip connections.

Recent advances in remote sensing show that combining complementary information from different sensing modalities can significantly improve segmentation performance. For instance, integrating spectral data from multispectral or NIR sensors with structural cues from LiDAR can help models better distinguish vegetation types and boundaries under varying environmental conditions [6]. Beyond simply merging inputs, some works aim to explicitly transfer knowledge across heterogeneous sensors, reducing the need for all modalities during deployment. For example, Huo et al. [7] proposed C2KD, which distills information from a multi-modal teacher to a single-modal student, effectively narrowing the modality gap. Inspired by these developments, our study focuses on achieving high segmentation accuracy while relying on fewer input modalities, using knowledge distillation as the transfer mechanism.

B. Knowledge Distillation for Semantic Segmentation

Knowledge distillation (KD) is a model compression technique in which a smaller, more efficient student network is trained to replicate the behavior of a larger, high-performing teacher network. The goal is typically to transfer the teacher's generalization ability to the student so that the student can

achieve comparable performance with fewer parameters, reduced inference time, or simplified input requirements. In this work, however, our primary focus is on reducing input modality requirements rather than minimizing model size, leveraging KD to transfer knowledge from a multi-modal teacher to a single-modal student. Originally introduced by Hinton et al. [8], KD has since been widely applied in classification, detection, and segmentation tasks.

In semantic segmentation, KD provides dense, pixel-level supervisory signals that go beyond standard one-hot ground-truth labels. The teacher produces *soft labels* that represent class probability distributions for each pixel, capturing inter-class relationships and subtle boundary information. For example, if a pixel lies between vegetation and road in the ground truth, the teacher may assign 0.7 probability to “vegetation” and 0.3 to “road” encouraging the student to learn richer decision boundaries.

KD can be categorized into three main types [9]:

- *Response-based KD* – transfers the teacher’s output logits or probabilities (soft targets) directly to the student.
- *Feature-based KD* – aligns intermediate feature maps between teacher and student to match internal representations.
- *Relation-based KD* – preserves structural relationships, such as similarities between features, attention maps, or spatial correlations.

In this work, we adopt response-based KD because our goal is to transfer final prediction behavior from a multi-modal teacher to a single-modal student without requiring architectural alignment or access to intermediate feature maps. Response-based KD is straightforward to implement, requires no changes to the backbone, and is well suited for modality-reduction scenarios where only the final output space is shared between teacher and student.

The general KD loss function is a weighted combination of the hard target loss (against ground-truth labels) and soft target loss (against teacher outputs):

$$L_{\text{total}} = \alpha L_{\text{hard}} + (1 - \alpha) L_{\text{soft}} \quad (1)$$

where $\alpha \in [0,1]$ balances the influence of ground truth and teacher guidance. Typically, L_{hard} is a categorical cross-entropy loss and L_{soft} is a Kullback-Leibler (KL) divergence between the temperature-scaled teacher and student probability distributions:

$$L_{\text{soft}} = T^2 \text{KL}(p_s^T \parallel p_t^T) \quad (2)$$

where:

- p_s^T - the student model’s predicted class probability distribution for each pixel, obtained after applying the softmax function with temperature T .
- p_t^T - the teacher model’s predicted class probability distribution for each pixel, also softened using the same temperature T .
- T - the distillation temperature, a scalar $T > 1$ that scales the logits before the softmax, producing a

smoother probability distribution that exposes relative probabilities among non-maximum classes.

- KL - the Kullback–Leibler divergence, which measures how one probability distribution diverges from another. In this context, it quantifies how close the student’s softened predictions are to the teacher’s softened predictions.
- T^2 - a scaling factor recommended by Hinton et al. [8] to maintain the magnitude of gradients when using higher temperatures, ensuring balanced contributions to the total loss.

Here, using $T > 1$ softens the output probabilities, revealing inter-class similarities that may be hidden in one-hot labels and providing more informative gradients during training.

III. PROPOSED METHOD

A. Dataset

We used an urban subset of the multi-source vegetation dataset assembled by the Korea National Geographic Information Institute (NGII) and NASA’s GEDI program in 2022. The dataset consists of 132,000 ortho-rectified GeoTIFF tiles (81.8 % forest and 18.2 % urban) with a 10 cm ground-sampling distance. Each 512×512 px tile has co-registered RGB, NIR and LiDAR-derived canopy-height channels and pixel-level masks for six land-cover classes: *Coniferous Trees, Broadleaf Trees, Non-Forest Area, Street Trees, Grassland and Shrubs*.

B. Framework Overview

Fig. 2 presents our three-stage knowledge distillation pipeline, which takes RGB, NIR, and LiDAR data as inputs and produces a compact RGB-only segmentation model.

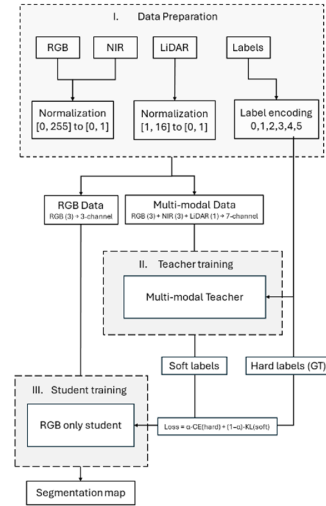


Fig. 2. Framework diagram of the training pipeline, illustrating data normalization and encoding, multi-modal U-Net teacher training, and RGB-only student distillation using a combined cross-entropy and KL-divergence loss.

- 1) *Stage I*: We utilized the original ortho-rectified 512×512 px GeoTIFF tiles without cropping. Each RGB and NIR channel was normalized from $[0, 255]$ to $[0, 1]$, while LiDAR canopy-height values in $[1, 16]$ m were

normalized to [0, 1]. Pixel-level labels were integer-encoded into six classes. We ensured co-registration across RGB, NIR, and LiDAR by matching the shared image IDs embedded in file names. The dataset exhibits class imbalance, with the Non-Forest Area class dominating (>70% of pixels) and minority classes such as Street Trees and Shrubs each comprising <3% of pixels.

- 2) *Stage II*: The RGB, NIR, and LiDAR modalities were concatenated into a 7-channel input tensor for the multi-modal U-Net teacher. Training was performed for a fixed 100 epochs without early stopping. No pre-trained weights were used, all parameters were initialized via Xavier initialization. The model was optimized with Adam (learning rate 1×10^{-4} , batch size 8) using cross-entropy loss against the hard ground-truth labels.
- 3) *Stage III*: The student model used only RGB inputs (3 channels). During training, it minimized a weighted sum of categorical cross-entropy on the hard ground-truth masks and KL divergence on the teacher's temperature-softened outputs. We set the loss weight coefficient $\alpha = 0.5$ to equally balance the influence of hard labels and soft teacher outputs, and the distillation temperature $T = 3$ following Hinton et al. [8]. This produced sufficiently softened probability distributions to provide stable gradients and reveal inter-class similarities without making the distribution overly uniform. All layers were trained from scratch with the same optimizer settings as the teacher. No data augmentation, class re-weighting, or label smoothing was applied, ensuring a controlled baseline. The result is a lightweight segmentation model that achieves near-teacher accuracy while requiring only single-modality input.

C. Network Architecture

All models teacher, distilled student, and both baselines share the same U-Net architecture to isolate the effect of input modalities. The first convolutional layer differs only by input channels: the teacher uses 7 channels (3 RGB, 3 NIR, 1 LiDAR), the RGB-only baseline and student use 3 channels (RGB), and the NIR-only baseline uses 3 NIR channels.

The U-Net architecture, as illustrated in Fig. 1, consists of an encoder path that captures context and a symmetric decoder path that enables precise localization. The encoder is composed of four blocks, each containing two 3x3 convolutional layers followed by a ReLU activation function and a 2x2 max pooling operation for downsampling. The decoder path also has four blocks, each consisting of a 2x2 up-convolution, a concatenation with the corresponding feature map from the encoder path, and two 3x3 convolutional layers with ReLU activation. The final layer is a 1x1 convolution that maps the feature channels to the desired number of classes.

D. Implementation Details

We compared four U-Net variants under identical training settings. Inputs were normalized as described in Fig. 2, and no data augmentation was applied. This ensures that any performance differences stem solely from modality fusion and the distillation process rather than other training tricks. The distillation temperature was $T = 3$ and the loss-weight coefficient $\alpha = 0.5$, matching the Stage III settings. These values were not further tuned to maintain a clear, reproducible baseline.

IV. EXPERIMENT RESULTS AND ANALYSIS

A. Evaluation Metrics

We report mean IoU (mIoU) across all classes, individual class IoUs, and the macro-averaged F1-score:

$$\text{mIoU} = \frac{1}{C} \sum_{i=1}^C \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \quad (3)$$

Per-class IoU highlights class-wise performance; the per class F1-score (the harmonic mean of precision and recall) is useful for evaluating imbalanced classes:

$$\text{F1} = \frac{2\text{TP}_i}{2\text{TP}_i + \text{FP}_i + \text{FN}_i} \quad (4)$$

B. Overall Performance Comparison

Table I reports overall mIoU and macro-averaged F1-score. The multi-modal teacher sets the upper bound with mIoU = 0.7132 and F1 = 0.8253. The RGB-only baseline scores 0.4091 mIoU. The distilled student jumps to 0.6002 mIoU and 0.7342 F1, closing 0.1911 of the 0.3041 gap to the teacher and trailing by just 0.113 mIoU. The NIR-only baseline sits between them at 0.5044 mIoU.

TABLE I. OVERALL MIOU PERFORMANCE COMPARISON.

Model	Method	mIoU	Macro F1-Score
Teacher	Supervised	0.7132	0.8253
Student (Distilled)	Distilled	0.6002	0.7342
Baseline (RGB-only)	Supervised	0.4091	0.5206
Baseline (NIR-only)	Supervised	0.5044	0.6366

Table II and Table III detail per-class IoU and F1-score. Distillation yields the largest gains on challenging classes. For example, the student's IoU for Shrubs increased from 0.0491 (RGB-only) to 0.3671 (distilled), a $7.5\times$ improvement, and F1 rises from 0.0935 to 0.5371. Coniferous Trees' IoU climbs from 0.1447 to 0.4725, and Street Trees from 0.2547 to 0.4788. These improvements show that the distilled student successfully captures the teacher's fused multi-modal knowledge.

TABLE II. PER-CLASS MIOU PERFORMANCE COMPARISON.

Class	Teacher	Student	RGB-only	NIR-only
Coniferous Trees	0.5717	0.4725	0.1447	0.2915
Broadleaf Trees	0.7500	0.6488	0.5120	0.5880
Non-Forest Area	0.9512	0.9164	0.8928	0.9259
Street Trees	0.5945	0.4788	0.2547	0.3700
Grassland	0.8111	0.7177	0.6016	0.6453
Shrubs	0.6008	0.3671	0.0491	0.2059

TABLE III. PER-CLASS F1-SCORE COMPARISON.

Class	Teacher	Student	RGB-only	NIR-only
Coniferous Trees	0.7275	0.6418	0.2519	0.4514
Broadleaf Trees	0.8572	0.7870	0.6773	0.7405
Non-Forest Area	0.9750	0.9564	0.9433	0.9615
Street Trees	0.7457	0.6475	0.4061	0.5402
Grassland	0.8957	0.8356	0.7512	0.7844
Shrubs	0.7506	0.5371	0.0935	0.3414

C. Discussion

The distilled student closes much of the gap between single-modal RGB and multi-modal teacher performance,

trailing the teacher by only 0.113 mIoU. The student's score of 0.6002 mIoU represents an improvement of 0.1911 over the RGB baseline, which corresponds to recovering about 62.8 % of the performance that would otherwise be lost without the NIR and LiDAR modalities (i.e., the student regains roughly two-thirds of the accuracy gap between the baseline and the teacher).

This result is notable because the student operates with less than half the input information of the teacher and without training enhancements such as pretraining, augmentation, or complex loss schedules. Compared to prior studies in semantic segmentation distillation, our gap is on par with or smaller than many same-modality setups. For example, Fan et al. [10] report a 0.0326 mIoU boost with Af-DCD, Wu et al. [11] achieve a 0.014 mIoU gain with Spirit Distillation, and Liu et al. [12] obtain up to 0.0378 mIoU improvement using Structured Knowledge Distillation. Our results are achieved in a more challenging cross-modal setting, where only the final prediction space is shared between teacher and student.

Per-class results reveal the strongest relative gains in challenging minority categories, particularly *Shrubs* and *Street Trees*, which typically suffer from low recall in RGB-only model. These improvements suggest that the multi-modal teacher's structural and spectral cues are successfully distilled into the RGB-only student, helping it resolve ambiguities and refine class boundaries. At the same time, performance in dominant classes such as *Non-Forest Area* remains stable, indicating that KD improves minority-class accuracy without degrading major categories.

Overall, the findings confirm that even a straightforward response-based KD approach can effectively compress multi-modal knowledge into a single-modal network, offering a competitive balance between accuracy, efficiency, and sensor requirements.

V. LIMITATIONS AND FUTURE WORK

Despite promising results, our approach has several limitations. First, the student's mIoU remains 0.113 below the teacher's, having recovered only 0.1911 mIoU of the original 0.3041 mIoU gap. This suggests that more sophisticated distillation losses (e.g., adaptive inter-class similarity distillation by Mansourian et al. [13]) and novel architectural variants could further narrow this gap. Second, response-based KD may not fully bridge the modality gap between RGB and LiDAR; exploring cross-modal frameworks such as C2KD by Huo et al. [7] could enhance transfer effectiveness. Third, our experiments use a single, custom urban dataset; validating the method on diverse public benchmarks such as OpenEarthMap [14] and LandCoverNet [15] is essential to demonstrate generalization across different sensors, environments, and regions. Finally, the teacher model was trained under basic settings; adopting advanced backbones like Swin-UNet and richer augmentation policies could raise the performance ceiling and yield even stronger students.

VI. CONCLUSION

We presented a simple cross-modal knowledge distillation framework to reduce input requirements for urban vegetation segmentation. A multi-modal U-Net trained on RGB, NIR and LiDAR achieved 0.7132 mIoU, while our distilled RGB-only

student reached 0.6002 mIoU outperforming RGB-only (0.4091 mIoU) and NIR-only (0.5044 mIoU) baselines by significant margins. Per-class results confirmed the student's ability to leverage multi-modal cues, with especially strong gains on difficult classes. By transferring fused spectral and structural knowledge into a single-modal model, we simplify the inference pipeline and reduce sensor and processing costs with only a modest accuracy drop. These findings suggest that even basic KD can effectively bridge modality gaps in remote sensing, opening the door to more scalable, cost-effective deployment of high-performance segmentation models.

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